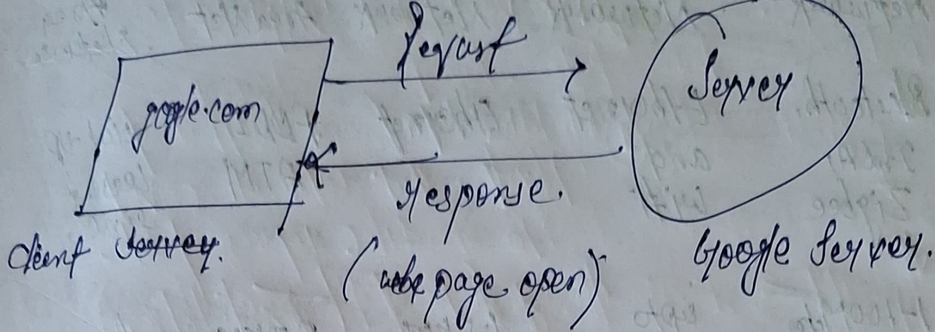
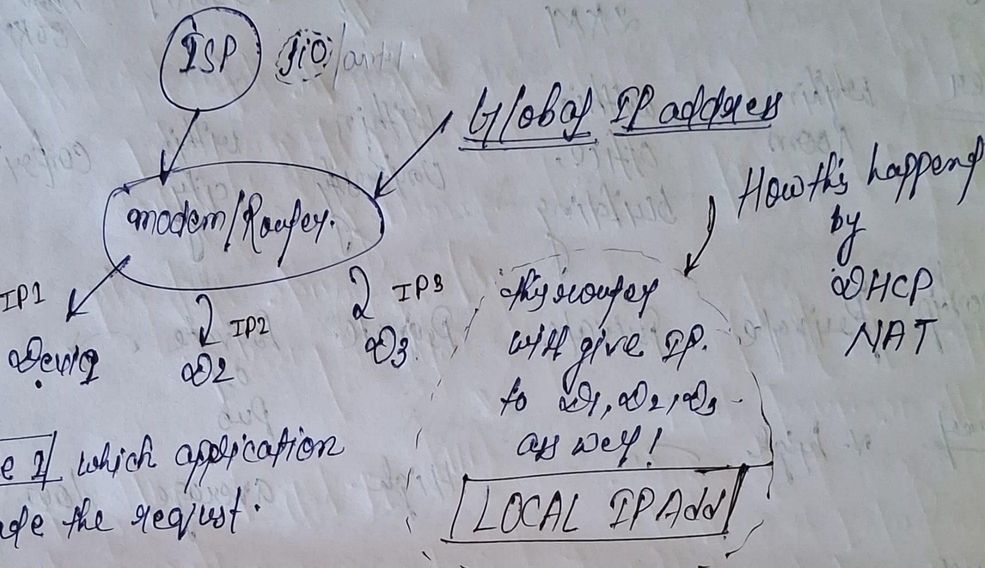
Computer Network

Date- 06/06/2026.
+ Kanal Kushwaha.





In device 1 which application is made the request.

- IP Address decide which device to send the data.
  but how do we get which application to send the data in particular device
  We use by PORT

Port → 16 b → 2^{16} ≈ 65,000.

HTTP = 80 | 0 — 1023. (reserved port)
MongoDB - 27017. | 1024 - 49152 (application)
SQL = 1433 | remaining u can use.

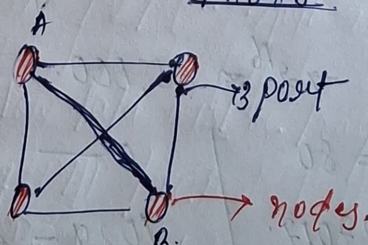
1 mbps = 1000000 bits/sec.
1 gbps = 10^9 bits/sec.
1 Kbps = 1000 bits/sec

	PAN	LAN	CAN	MAN	WAN.
	Personal Area Network	local Area Network	campus Area Network	Metropolitan Area Net	wide Area Networ
Technology	Bluetooth, IrDA, Zigbee	Ethernet and wifi	Ethernet	FDDI ATM	dial-up leased line.
Range	1-100 Meter	upto 2 KM	1.5 KM	5-50 KM	Above 50 KM.
Area	within Room	within office, building	within University	within city	country.
Ownership	Private	Private	Private.	Private or Pub	→
Speed	V. high.	→ "	High	average	low.
Error Rate & cost	very low	low	avrage.	High	V. High

Topology.

① Mesh topology :

4 nodes



No of cables - $\frac{n \times (n-1)}{2}$

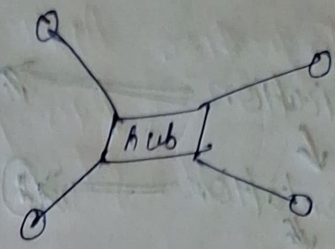
No of ports - $(n-1)(\times n) \rightarrow$ total ports.

Reliablity - ↑

cost - ↑

Security - Yes, if A and B transferring the data other have no clue.

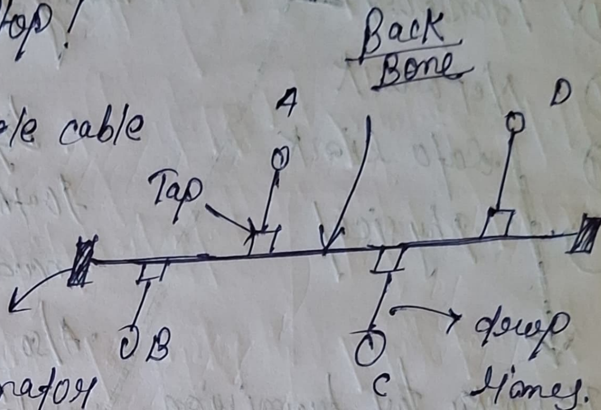
(II) Hub / Star : All nodes connect to central Hub/Switch.



- Single point failure → in case some Problem comes in hub then the four cannot communicat with each other.

Whole LAN network will stop!

(III) Bus : All nodes on single cable



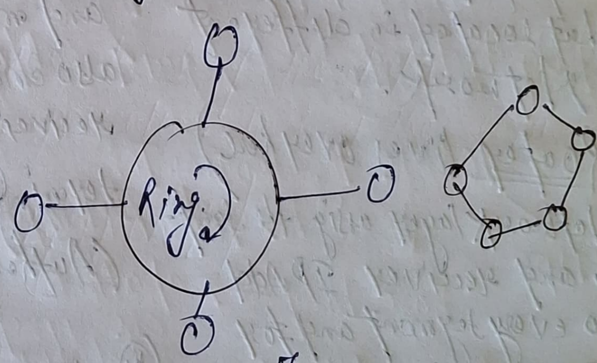
No of cables : n+1

n → Total node, 1 → Back Bone

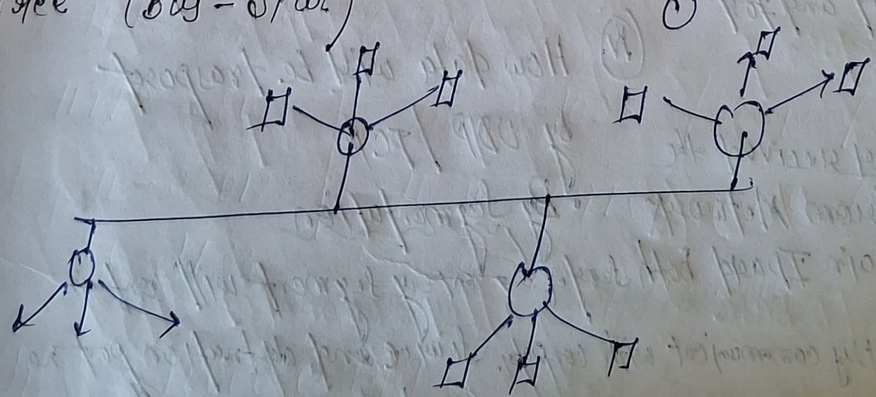
Port: 2*n

- Single point failure → No reliabilty.
- Security → less HOT

(IV) Ring : Same as Bus.



(V) Tree (Bus - Star)



OSI Model:

(I) Application → (I) implemented in Software.
(II) Presentation → (II) It take data from the application layer, that data it convert to machine representable binary format. (Translation)

Before the data transmitted further it goes under encoding → encryption

· Also provide Abstraction.

· Data compressed.

(III) Session

(III) It helps to setting up and managing the connections also enable the sending / receiving the data, followed by termination to connected session. (Authentication → Authorisation.)

(IV) Transport

(IV) How data will be transport by UDP / TCP

· By Segmentation
↳ every segment will contain source and destination port no. and sequence number.

· flow control
transport layer control the amount of data being transport

· Error control
data lost or gain about?

(V) Network

It works for transmission of the received data segment from one computer to another. that located in different Network.

(Router lives over here)

· Network layer assign the sender and receiver IP Add to every segment and for IP Packet.

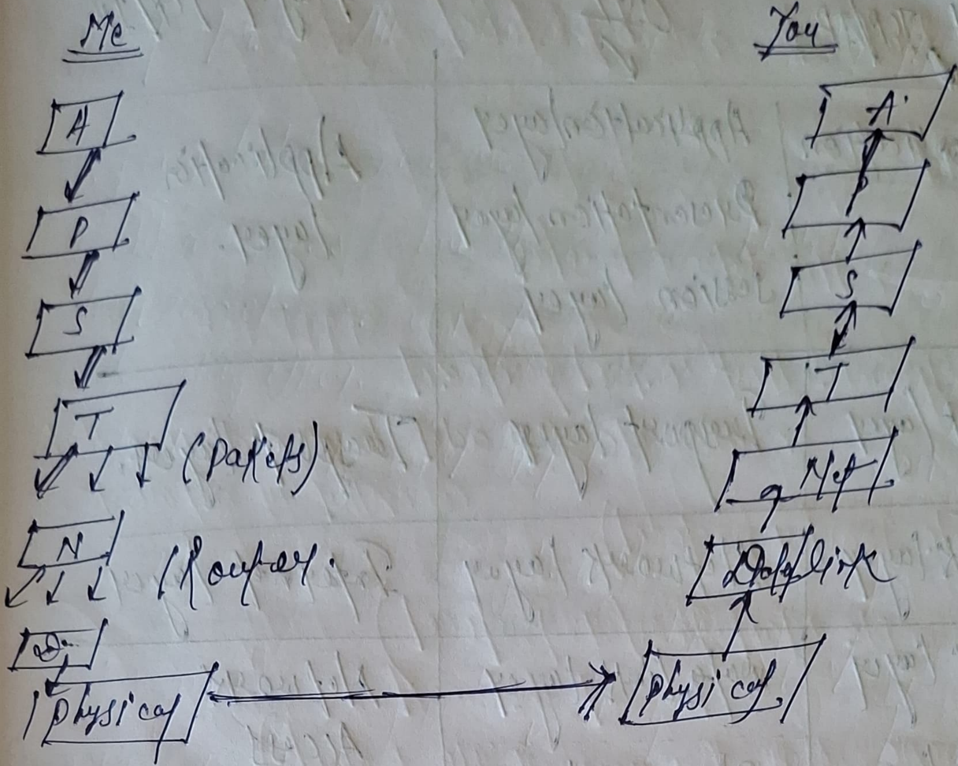
(VI) Data link

Data link layer received the data packet from Network this packet contain IP add both sender and receiver.

· It allow directly communicat with computer.

(VII) Physical

physical layer received the signal converts it into bits pass it to data link layer as a frame and it will move to higher higher



TCP/IP Models

(Transmission control protocol / Internet protocol)
Reference Model is a networking model used for communication over Internet. It define how data is transmitted from one device to another.

Application → Transport → Network → data link → physical.

OSI Model	TCP/IP Models
· It has 7 layers	· 4 layers.
· Developed by (ISO)	· DARPA
· It is a reference Model	· It is a protocol suite and reference model
· Layers are clearly separated	· Some layers are combined.
· Session + Presentation layer separated.	· Merged to Application layer.
· less commonly used in Real Network	· Widely used on the Internet.

By Firozenbook.

5 layer TCP/IP	OSI	4 layer TCP/IP.
Application	Application layer Presentation layer Session layer	Application layer.
Transport layer	Trasport layer	Trasport layer.
Network layer	Network layer	Internet layer.
Datalink layer Physical layer	Data link layer Physical layer.	Network Access layer.

Types of cables.

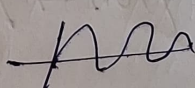
(I) Unshielded twisted pair cable (LAN, telephone)

10 Base T, 100 Bas T

10 Mbps

100 mt (meter)

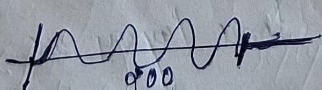
Base band → at a time one signal can travel.



Broad band → at a time more signal can travel

(II) coaxial cable. (

10 Base 2 → after 200 meter signal attenuate.



(III) Fiber optic → 100 Bas Fx